

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0459

Koch Oil 100 kV
March 26, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0459. The Interconnection Customer proposes a 263.0 MW natural gas combined cycle facility interconnecting at the Koch Oil 100 kV substation in the Ohio area. The request was received on April 10, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 276 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,396,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Maclean – Broken Bow 100 KV
Contingency: Broken Bow – Fort Hall 100 KV
Loading: 179.3% (Rating: 101 MVA)
DFAX: 41.8%
Pre-project loading: 70.3%
2. Facility: Maclean – Ft Hall 100 KV line
Contingency: Fort Hall – Broken Bow 100 KV
Loading: 139.4% (Rating: 118 MVA)
DFAX: 41.8%
Pre-project loading: 46.5%
3. Facility: Deer Park – Corbell 100 KV Line
Contingency: J R Whiting 230-1 kV TX
Loading: 137.6% (Rating: 326 MVA)
DFAX: 83.0%
Pre-project loading: 70.6%
4. Facility: Broken Bow – Corbell 100 KV
Contingency: J R Whiting 230/1kV XFMR
Loading: 137.6% (Rating: 326 MVA)
DFAX: 83.0%
Pre-project loading: 70.6%
5. Facility: Deer Park Substation – Koch Oil 100 KV L
Contingency: J R Whiting 230/1 kV Auto
Loading: 135.1% (Rating: 352 MVA)
DFAX: 83.0%
Pre-project loading: 73.0%

6. Facility: Ft Hall – Tracy 100 KV
Contingency: Bayport Sw Sta – Firstenergy Bay Shore 230 KV
Loading: 127.9% (Rating: 141 MVA)
DFAX: 29.6%
Pre-project loading: 72.5%
7. Facility: Ft Hall – Broken Bow 100 KV
Contingency: Bayport Sw Sta – Firstenergy Bay Shore 230 KV line
Loading: 120.1% (Rating: 263 MVA)
DFAX: 42.4%
Pre-project loading: 77.8%
8. Facility: Big Creek 4 – Ft Hal 100 KV L
Contingency: Firstenergy Bay Shore – Bayport Sw Sta 230 KV
Loading: 113.9% (Rating: 188 MVA)
DFAX: 27.3%
Pre-project loading: 75.7%
9. Facility: Gray'S Flat – E Methuen 230 KV line
Contingency: E Methuen – Madeline 230 KV
Loading: 105.8% (Rating: 529 MVA)
DFAX: 7.6%
Pre-project loading: 102.0%
10. Facility: Doyle Energy Facility – Toll Rd 100 KV
Contingency: J R Whiting 230 kV/1 kV Auto
Loading: 103.0% (Rating: 183 MVA)
DFAX: 9.5%
Pre-project loading: 89.4%
11. Facility: Bonita Springs – Toll Road 100 KV line
Contingency: J R Whiting 230/1 kV Transformer Bank
Loading: 103.0% (Rating: 183 MVA)
DFAX: 9.5%
Pre-project loading: 89.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Maclean – Broken Bow 100 KV	Rebuild line with higher-capacity conductor	\$1,569,000	52% (\$815,000)
Maclean – Ft Hall 100 KV line	Rebuild line with higher-capacity conductor	\$3,318,000	52% (\$1,722,000)
Deer Park – Corbell 100 KV Line	Rebuild line with higher-capacity conductor	\$1,590,000	36% (\$576,000)
Broken Bow – Corbell 100 KV Line	Rebuild line with higher-capacity conductor	\$2,030,000	36% (\$735,000)
Deer Park Substation – Koch Oil 100 KV L	Rebuild line with higher-capacity conductor	\$2,727,000	36% (\$987,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ft Hall – Tracy 100 KV	Rebuild line with higher-capacity conductor	\$2,861,000	48% (\$1,369,000)
Broken Bow – Ft Hall 100 KV	Rebuild line with higher-capacity conductor	\$1,258,000	46% (\$582,000)
Fort Hall – Big Crk4 100 KV L	Rebuild line with higher-capacity conductor	\$2,650,000	45% (\$1,184,000)
Gray'S Flat – East Methuen Substation 230 KV L	Reconductor line	\$15,282,000	6% (\$983,000)
Doyle Energy Facility – Toll Rd 100 KV	Reconductor line	\$1,014,000	11% (\$113,000)
Bonita Springs – Toll Rd 100 KV	Reconductor line	\$2,948,000	11% (\$330,000)
	Total allocated to project		\$9,396,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.